

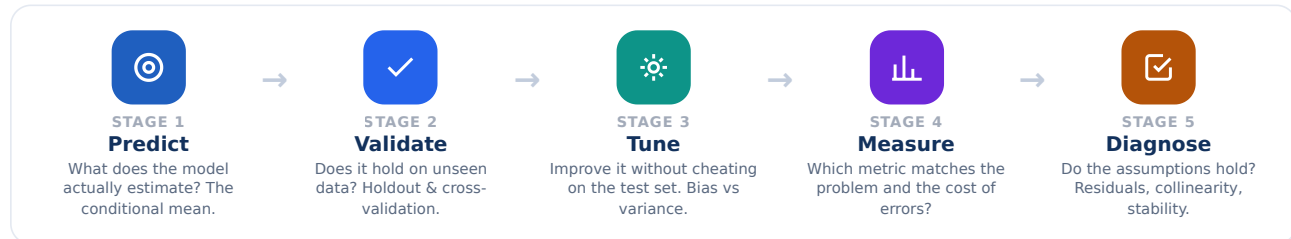
THE WHOLE JOB IN ONE LINE  
 “Would this model still be right on a slab it has never seen before?”

# The Evaluation Workflow

From “what does it predict?” to “can I trust it?”

Evaluation is not one number at the end — it is five questions asked in order. Skip one and you can ship a model that looks brilliant and fails in production.

## THE FIVE-STAGE WORKFLOW



### The one question behind it all

Every stage is a way of answering one thing honestly: **how well does this model do on data it has never seen?** Training performance is a diagnostic, not an achievement.

Fit  $\neq$  Generalization    Accuracy  $\neq$  Validity  
 Software ran  $\neq$  Answer is right

### Three verdicts you're trying to avoid



## GOLDEN RULES OF HONEST EVALUATION

- 1 Judge on data the model never saw.** The cross-validation score is the truth; the training score is just a diagnostic.
- 2 Match the metric to the decision,** not to habit. The cost of each error picks the metric.
- 3 Keep one test set untouched** until the very end — tune on cross-validation only.
- 4 A model can predict well and still break assumptions** — read metrics *and* diagnostics.
- 5 Prevent group leakage:** the same slab in train and test inflates every score.
- 6 Never trust a single number** — read fit and generalization together, always.

## READING THE RESULT IN THREE MOVES



### KEY TAKEAWAY

Evaluation is a **sequence of honest questions**, not a final score. Ask all five, in order, every time.

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